FLEXIBLE COMPUTING LOADS, ELECTRICITY-GRID RESILIENCE, AND CARBON EMISSIONS: A GENERAL THEORY OF WHEN MINING AND AI TRAINING REDUCE EMISSIONS

TEAM A1 (WORKSTREAM A: OPTIMAL CONTRACTING THEORY AND QUANTITATIVE VALIDATION)

Abstract. Intensive, interruptible computing demands—Bitcoin mining and, imperfectly, AI training—can be dispatched in real time. We ask when introducing such a *flexible computing load* (FCL) into a regulated grid *reduces* carbon emissions. The program changes the utility's optimal mix of renewable (K_R), fossil (K_F), and computing (K_M) capacity, setting up two opposing forces: a distribution-free *utilization effect* (computing burns spare fossil, raising emissions) and a *capacity-substitution effect* (computing revenue $R > c$ raises the shadow value of renewables, so $K_R^* > K_R^\circ$, lowering emissions). Our contributions: (i) we prove the capacity-substitution channel $K_R^* > K_R^\circ$ for *any* residual-demand distribution satisfying a transparent shift-ratio condition plus interior regularity; (ii) we give the exact net-carbon inequality and a clean sufficient condition; (iii) we identify a closed-form *viability threshold* on the renewable cost, with a rigorous general large-cost anchor and an explicit power-law window; and (iv) at fixed equilibrium capacities we reduce the carbon effect to a two-mean affine form, bracket it by closed-form bounds, and identify the uniform as the least-favorable conditional shape. Every analytical formula is verified to machine precision, and the distribution-free claims are confirmed across eight distribution families. We are explicit about what remains conditional: the fully general, finite-optimum carbon guarantee is proved for the power-law family and conjectured beyond it.

1. Introduction

Bitcoin mining consumes electricity on a national scale, and AI training is on a similar trajectory. Both are unusual loads: they can be ramped down in seconds at little cost, so a grid operator can treat them as a real-time *demand-response* resource. A provocative possibility follows—that adding such a load could *lower* carbon emissions, by making renewable investment more profitable (the computing load monetizes otherwise-curtailed renewable surplus) even as it also burns some fossil fuel when it runs on spare thermal capacity. Whether the net effect is lower or higher emissions is the question of this paper.

We study a regulated utility that chooses renewable capacity K_R, dispatchable fossil backup K_F, and flexible-computing-load capacity K_M to maximize expected profit, and dispatches the computing load by an optimal real-time rule. Introducing the program sets up two forces on carbon. The **utilization effect** is that computing run on spare fossil capacity strictly raises fossil output—this always pushes emissions up and is distribution-free. The **capacity-substitution effect** is that the computing margin $R - c > 0$ raises the shadow value of renewables, so the utility builds *more* renewables ($K_R^* > K_R^\circ$), which shifts residual demand toward surplus and pushes emissions down. The carbon question is which force wins.

Contributions and roadmap. This integrated draft is the joint output of four roles. §2 sets up the model and the optimal dispatch contract. §3 proves the capacity-substitution channel $K_R^* > K_R^\circ$ in general (Role 1). §4 gives the exact net-carbon condition and its distribution-free reading (Role 1). §5 establishes the renewable-cost viability threshold (Role 1). §6 maps the regimes, completes the small-FCL case, and pressure-tests the maintained assumptions (Role 4). §7 reports the solver, the

Date: Integrated working draft.

machine-precision audit, and the comparative statics across eight distribution families (Roles 2–3). §8 states honestly what is proved versus conditional, and the path to a complete theory.
Scope. This is the *general theory and numerical methods* contribution. Empirical calibration to real grids, the reliability/welfare and social-versus-private analysis, and the generalization to a flexibility continuum (storage, partially-flexible AI, multi-grid markets, mining dynamics) are companion efforts and are out of scope here.

2. Model and the optimal dispatch contract

A regulated utility serves a representative period. The state is *residual demand*

$$\tilde{x} := \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}], \tag{1}$$

where $\tilde{D} \in [0, \bar{D}]$ is stochastic consumer demand and $\tilde{S} \in [0,1]$ the renewable capacity factor; $F(x; K_R)$ is the CDF with density f, extended by $F \equiv 0$ below $-K_R$ and $F \equiv 1$ above $\bar{D}$. More renewables shift mass left: $F_{K_R} := \partial F/\partial K_R \geq 0$. Per-MWh: computing revenue R, fossil cost c, retail price p, blackout penalty β. Amortized per-MW capital: C_R, C_F, C_M.

Assumption 1 (Regularity; maintained throughout)**.** $R > c + C_M, \quad R < c + C_F + C_M, \quad \beta + p > c + C_F, \quad \beta + p > R.$

A *demand-response contract* is a dispatch policy $d : [-K_R, \bar{D}] \to [0, K_M]$. Computing can only run on power physically available (spare fossil plus renewable surplus), $a(x) := (K_F - x)_+$.

Lemma 1 (Optimal contract; Role 1)**.** *For any F and any capacities, the unique optimal served dispatch is the renewable-first threshold rule $d^*(x) = \min\{(K_F - x)_+, K_M\}$: full computing on deep surplus, exact gap-filling on $(K_F - K_M, K_F)$, zero in scarcity.*

Substituting d^* and integrating by parts gives the profit

$$\Pi = p\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F-K_M}(1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}}(1-F)\,dx - C_FK_F - C_RK_R - C_MK_M, \tag{2}$$

with first-order conditions (valid in both regimes; Roles 1,4)

$$\frac{\partial\Pi}{\partial K_M} = (R-c)F(K_F - K_M) + cF(-K_M) - C_M, \tag{3}$$

$$\frac{\partial\Pi}{\partial K_F} = R[F(K_F) - F(K_F - K_M)] - c[1 - F(K_F - K_M)] + (\beta+p)[1 - F(K_F)] - C_F, \tag{4}$$

$$\frac{\partial\Pi}{\partial K_R} = R\int_{K_F-K_M}^{K_F} F_{K_R} + c\int_{-K_M}^{K_F-K_M} F_{K_R} + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_R} - C_R. \tag{5}$$

Lemma 2 (Distribution-free cutoffs; Roles 1,4)**.** *The no-FCL threshold satisfies $F(K_F^\circ; K_R^\circ) = \varphi^\circ := (\beta + p - c - C_F)/(\beta + p - c)$, independent of F. In the large-FCL regime ($F(-K_M^*) = 0$), also $F(K_F^*; K_R^*) = (\beta + p - c - C_F - C_M)/(\beta + p - R)$ and $F(K_F^* - K_M^*; K_R^*) = C_M/(R - c)$. Hence the adoption threshold $\bar{C}_M = (R - c)\varphi^\circ$.*

3. The capacity-substitution channel

The carbon result rests on $K_R^* > K_R^\circ$: the program builds more renewables. The driver is visible in (5): on the dispatch band $(K_F - K_M, K_F)$ the shadow value of renewables is weighted by R, whereas without computing it is weighted by $c < R$.

Condition 1 (Shift ratio)**.** $g(x) := F_{K_R}(x)/f(x) = \mathbb{E}[\tilde{S} \mid \tilde{x} = x]$ is non-increasing on the dispatch window at each point of the adoption path.

Condition 2 (Regularity)**.** Along the adoption path the program has an interior optimum at which the Hessian $\mathcal{H}$ of Π is nonsingular (hence $\prec 0$).

Theorem 1 (Capacity-substitution channel; Role 1)**.** *Under Conditions 1–2, renewable capacity is strictly higher with the program: $K_R^* > K_R^\circ$ whenever $C_M < \bar{C}_M$. The mechanism is the comparative static $dK_R^*/dC_M = (R-\beta+p)(R-c)\left[F_{K_R}(K_F^*)f(K_F^*-K_M^*) - F_{K_R}(K_F^*-K_M^*)f(K_F^*)\right]/\det\mathcal{H} \le 0$, which integrates from $\bar{C}_M$ (where $K_M^* = 0$, $K_R^* = K_R^\circ$) downward.*

The proof differentiates the FOC system implicitly; the cofactor sign reduces to $g(K_F^*) - g(K_F^* - K_M^*) \le 0$, exactly Condition 1. The obstruction to a one-line lattice argument is that Π is *submodular* in (K_R, K_F) while having increasing differences in K_M; the homotopy sidesteps it.

Corollary 1 (Power-law family; explains footnote-6)**.** *For $F(x; K_R) = ((x+K_R)/(\bar{D}+K_R))^\alpha$, $g(x) = (\bar{D}-x)/(\bar{D}+K_R)$ is affine and decreasing for* every *$\alpha > 0$, so $K_R^* > K_R^\circ$ holds throughout the family.*

Remark 1 (Reading; sharpness)*.* g is the conditional renewable factor, so Condition 1 says high-stress states are renewable-poor—a negative demand–supply association. The cofactor bracket flips sign with g, so an increasing g would reverse the channel; constructing a *realized* reversed-channel example is left to the robustness round (Role 3).

4. The net-carbon condition

Carbon equals c times expected fossil output. Integrating (2)'s fossil terms by parts (Role 1) gives the exact identities $\mathbb{E}[\text{fossil}^*] = K_F^* - \frac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\text{low}}^*)$ and $\mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ$, with $q^\circ = F(0; K_R^\circ)$ and conditional means $\mu_{\text{low}}^*, \mu_{\text{mid}}^\circ$.

Proposition 1 (Exact net-carbon inequality; Role 1)**.** *Carbon falls with the program iff*

$$\underbrace{\tfrac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\text{low}}^*)}_{\textit{capacity-substitution}} > \underbrace{K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ}_{\textit{net utilization}}. \tag{C}$$

The utilization effect is always positive and distribution-free; the capacity-substitution effect is signed by Theorem 1.

Corollary 2 (Sufficient conditions; Role 1)**.** *If $f(\cdot; K_R^*)$ is non-increasing on the FCL zone and $f(\cdot; K_R^\circ)$ non-decreasing on the no-FCL fossil zone (shape class $\mathcal{U}$; the uniform is the boundary), then (C) holds at least as strongly as under the uniform.*

A distribution-free reading (at fixed capacities; Role 1). Holding the equilibrium capacities and q° fixed, $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ is *affine* in $(\mu_{\text{low}}^*, \mu_{\text{mid}}^\circ)$, hence bracketed by closed-form bounds $K_F^\circ(1-\varphi^\circ) - K_F^* \le \Delta \le K_F^\circ - K_F^* + \frac{C_M}{R-c}(K_F^* - K_M^* + K_R^*)$ computable from cutoff data; and within $\mathcal{U}$ the uniform minimizes Δ (it maximizes μ_{low}^* and minimizes μ_{mid}°). These are fixed-capacity statements; turning the last into a genuine cross-F certificate requires signing the capacity covariation, which we leave open (§8).

5. The renewable-cost viability threshold

Write $\Delta(C_R)$ for the net fossil saving at the optimum given renewable cost C_R. Whether carbon falls is, in the end, about how cheap renewables are.

Theorem 2 (Viability threshold; Role 1)**.** (i, general) *For all C_R above a finite B_0, no renewables are built, and the program differs from the no-FCL grid only by the (positive) utilization effect, so $\Delta(C_R) = -U_0 < 0$: the carbon-reduction set is bounded above for every F.* (ii, power-law / knife-edge) *The thresholds are explicit, $C_R^\dagger = \frac{1}{2}[c + \frac{(R-C_M-c)^2}{R-c} + \frac{(R-c-C_F-C_M)^2}{\beta+p-R}]$ and $C_R^\circ = C_F^2/2(\beta+p-c)$ with $C_R^\circ < C_R^\dagger$; on $(C_R^\circ, C_R^\dagger)$ renewables are unbounded, $\mathbb{E}[\textit{fossil}^*] \to 0$, and carbon strictly falls. Hence $C_R^\dagger = \sup\{C_R : \Delta \ge 0\}$ is finite, and positive for the power-law family.*

The large-cost anchor (i) is fully general; the strict-fall window (ii) uses the knife-edge (constant renewable marginal value) of the power-law family. For a generic F with an interior renewable optimum, the set $\{\Delta \geq 0\}$ is bounded above by (i), but its *non-emptiness*—that carbon falls somewhere—is what Condition 1 delivers in the cases we can sign and is conjectured in general. Closing this is the keystone of the next round.

6. Regimes and robustness of the assumptions

(Role 4: theory hardening.)
Small-FCL regime. When $K_M < K_R$ the surplus term $cF(-K_M)$ in (3) does not vanish; the first computing unit is worth more, widening adoption to $\bar{C}_M^b = \bar{C}_M + c\,q^\circ > \bar{C}_M$ (paper eq. 62), verified exactly.
Regime map. The C_M-thresholds are strictly ordered,

$$\underbrace{R - c - C_F}_{C_M^{\mathrm{fc}}} < \underbrace{(R-c)\varphi^\circ}_{\bar{C}_M} < \underbrace{\bar{C}_M + c\,q^\circ}_{\bar{C}_M^b},$$

sorting parameter space into no adoption, small-FCL, large-FCL interior, and full coverage; the boundedness axis $C_R^\circ < C_R^\dagger$ of §5 composes with it into one statement.
Necessity of Assumption 1. $R > c + C_M$ guarantees an interior K_M ($C_M/(R-c) < 1$); $\beta + p > R$ keeps the blackout cutoff well-defined and gives consumers priority; $\beta + p > c + C_F$ makes fossil backup worth building. The remaining bound $R < c + C_F + C_M$ is needed only for the *interior* regime— relaxing it lands in the well-defined full-coverage corner, not a contradiction.

7. Numerical methods and comparative statics

(Role 2: solver & validation; Role 3: comparative statics.)
Solver and exactness audit (Role 2). An independent solver quadratures (2) and solves (3)–(5) with no closed forms assumed. It reproduces the uniform closed forms to machine precision and confirms the draft's algebra, including three corrections (a dropped factor c in the printed carbon equations, a large-FCL flat term, and that the printed knife-edge bracket equals $2\,\partial\Pi/\partial K_R$). An adversarial audit checks all 15 analytical formulas (gradients, cross-partials, cofactor, distribution-free cutoffs, fossil decomposition) against finite differences: **15/15** agree to 6–16 digits; the test suite passes **36/36**.
Comparative statics across distributions (Role 3). Across eight families (uniform; Beta$(0.5,3)$, $(2,5)$, $(2,2)$, $(5,2)$; lognormal; exponential; Pareto):

- the distribution-free profit and blackout halves of the adoption result hold **16/16** whenever $C_M < \bar{C}_M$;
- the capacity-substitution channel $K_R^* > K_R^\circ$ holds **32/32**;
- the utilization effect is positive in every scenario;
- with capacities frozen (the short-run/timing case), carbon weakly rises **96/96**, confirming the sign of that comparative static;
- carbon *strictly falls* in **2/32** of the general-model scenarios, concentrated at low C_R and right-skewed F—exactly the regime Theorem 2 predicts.

The early reading "carbon never falls" and the theory "carbon can fall" are reconciled (Role 4): the sweep sat above the buildout crossover where the utilization effect dominates; the reduction obtains below $C_R^\dagger$ where renewables are unbounded. The thin "2/32" is consistent with the result and is the empirical leg the next round will strengthen into one curated, finite-optimum narrative.

8. Scope of the contribution and open problems

What is *proved*: the optimal dispatch contract (Lemma 1); the distribution-free cutoffs (Lemma 2); the capacity-substitution channel $K_R^* > K_R^\circ$ under Conditions 1–2 (Theorem 1); the exact net-carbon inequality (C) (Proposition 1); the fixed-capacity distribution-free reading; the general large-cost anchor and the explicit power-law carbon window (Theorem 2); the regime map and the necessity pattern of Assumption 1. All are machine-verified and cross-checked across roles.

What remains *conditional or open*, and is the agenda for completion:

(1) **General-F, finite-optimum carbon theorem (keystone).** Prove that $\{\Delta \geq 0\}$ is non-empty for general F under a *primitive* shift-ratio condition plus a tail condition, with the strict fall occurring at a finite interior K_R^* (not the unbounded corner).

(2) **Existence/uniqueness and regularity from primitives.** Replace Condition 2's numerically-checked $\mathcal{H} \prec 0$ with a primitive condition on F (e.g. log-concavity) guaranteeing a unique interior optimum.

(3) **Cross-F robust certificate.** Sign the capacity covariation so the least-favorable-distribution statement holds across $\mathcal{U}$, not only at fixed capacities.

(4) **Positioning.** A related-work section on the methods used (monotone comparative statics, interruptible/demand-response economics, distributionally-robust analysis) and a bibliography.

The spine is in place and internally consistent; these four items take the paper from a strong working draft to a complete *Management Science* submission.

Contributions by role. Role 1 (general carbon theory): the contract, the channel, condition (C), the threshold, the distribution-free reading. Role 2 (solver & numerical validation): the independent solver, the 15/15 audit, the corrections. Role 3 (comparative statics): the eight-family confirmations and the carbon-sign map. Role 4 (theory hardening & regimes): the small-FCL case, the regime map, the assumption pressure-test, and the theory–numerics reconciliation.